

DIGITAL SAT — MATH

Theory Companion — Part 2

Advanced Math & Geometry / Trigonometry — Theory & Worked Problems

- 16 skills, one representative Hard/Very Hard worked problem per skill
- Advanced Math: Functions · Composite & Inverse · Transformations · Polynomials & Factoring · Poly Division & Rational Expressions · Quadratics · Radicals & Exponents · Exponential Functions · Radical & Rational Equations
- Geometry and Trigonometry: Angles & Triangles · Right Triangles & Trig · Radians & the Unit Circle · Circles · Volume & Surface Area
- Companion to the full 16-skill, 128-question Math Theory Study Companion — Greek terminology included for every skill

The 16 Skills

1. Functions — Notation, Domain, Range & Graphs
2. Composite & Inverse Functions
3. Function Transformations
4. Polynomial Operations & Special Products
5. Factoring Polynomials
6. Dividing Polynomials & Rational Expressions
7. Solving Quadratic Equations & the Discriminant
8. Quadratic Graphs, Roots & Word Problems
9. Radicals & Rational Exponents
10. Exponential Functions — Growth & Decay
11. Radical & Rational Equations
12. Angles, Parallel Lines & Triangle Fundamentals
13. Right Triangles & Trigonometry
14. Radians & the Unit Circle
15. Circles — Circumference, Arcs, Sectors, Equations & Theorems
16. Volume, Surface Area & Similarity of Solids



PART 1

Advanced Math

Functions & Notation · Composite & Inverse Functions · Transformations · Polynomial Operations & Factoring · Polynomial Division & Rational Expressions · Quadratic Equations, Graphs & Roots · Radicals, Rational Exponents & Exponential Functions · Radical & Rational Equations

Core Rules — In Detail

- Function Definition & Notation — $y = f(x)$ $f(x)$ is read 'f of x' — the output of function f when the input is x . Each input in the domain maps to exactly one output.
- Evaluating Functions — $f(a) =$ (substitute a for x) To evaluate f at a given input, substitute that input for the variable everywhere it appears, then simplify.
- The Vertical Line Test — A graph represents a function if and only if no vertical line intersects it more than once.
- Domain and Range — Domain = valid x -values. Range = resulting y -values. On a graph, the domain is read along the x -axis and the range along the y -axis.
- Domain Restrictions — Rational: denominator $\neq 0$. Even root: radicand ≥ 0 . Combine every restriction (multiple denominators, multiple roots) into one final domain statement.
- Piecewise Functions — $f(x) = \{\text{expr 1 if cond 1; expr 2 if cond 2; ...}\}$ Determine which condition the input satisfies, then use only the matching expression to evaluate.
- Rate of Change — Rate of change = $(f(x_2) - f(x_1)) / (x_2 - x_1)$ For a linear function this equals the slope; in context it tells you how much the output changes per one-unit increase in the input.
- Points on a Graph — Point (a, b) on $y = f(x)$ means $f(a) = b$. To find x such that $f(x) = k$, locate where the horizontal line $y = k$ crosses the graph — there may be more than one such x .

Watch Out: $f(a) = b$ means 'when the input is a , the output is b ' — students often flip which number is the input and which is the output when a word problem gives function notation. ⚠ For domain questions with more than one restriction (a root AND a denominator, for example), find each restriction separately and then combine them — don't stop after the first one.

Let $f(x) = 3x - k$ and $g(x) = x^2 + 2x$, where k is a positive integer. If $g(f(1)) = 24$, what is the value of $f(k)$?

Formula / Rule Used: Function composition: substitute $f(1)$ into g , then solve the resulting equation for the unknown constant.

Work through it, then advance for the full step-by-step solution.

Step 1: Compute $f(1)$ in terms of k : $f(1) = 3(1) - k = 3 - k$.

Step 2: Substitute this into g : $g(3 - k) = (3 - k)^2 + 2(3 - k) = 24$.

Step 3: Let $u = 3 - k$ to simplify: $u^2 + 2u - 24 = 0$.

Step 4: Factor the quadratic: $(u + 6)(u - 4) = 0$, so $u = -6$ or $u = 4$.

Step 5: Convert back to k using $k = 3 - u$: if $u = 4$, $k = -1$; if $u = -6$, $k = 9$.

Step 6: Since the problem states k is a positive integer, reject $k = -1$ and keep $k = 9$.

Step 7: So $f(x) = 3x - 9$. Compute $f(k) = f(9) = 3(9) - 9 = 27 - 9 = 18$.

Answer: $f(k) = 18$

Tip: When a composition gives a quadratic in disguise, solve for the inner variable first, then translate every root back to the original constant before applying any stated restriction (like 'positive integer') to throw out invalid roots.

Trap: A student who finds both roots of the quadratic but forgets to check the 'k is a positive integer' condition may keep $k = -1$, which produces a completely different (and wrong) final answer.

Tempting Wrong Answers — and Exactly What Went Wrong

$$f(k) = -2$$

Used the rejected root $u = 4$ (which gives $k = -1$) instead of checking the positive-integer condition, then computed $f(-1) = 3(-1) - (-1) = -2$ with that invalid k .

$$f(k) = 24$$

Stopped after finding $g(f(1)) = 24$ and reported that given value instead of continuing on to solve for k and then evaluate $f(k)$.

$$f(k) = 9$$

Correctly solved for $k = 9$ but then reported k itself as the final answer instead of substituting it back into $f(x) = 3x - 9$.

Shortcut: There's no shortcut around setting up and solving the quadratic in u , but working with $u = 3 - k$ instead of expanding everything in terms of k keeps the algebra one step shorter.

Desmos Tip: Graph $y = (3 - x)^2 + 2(3 - x) - 24$ (as a function of k) and find its zeros; the two x -intercepts are -1 and 9 , making it easy to see which one is the positive integer.

Core Rules — In Detail

- Composite Function Notation — $(f \circ g)(x) = f(g(x))$ This means: apply g first, then apply f to that result. Order matters — $f(g(x))$ is generally not the same as $g(f(x))$.
- Method 1 — Work Inside Out — Evaluate the inner function first at the given number, then plug that numeric result into the outer function.
- Method 2 — Find the Composite Expression First — Substitute the entire expression for $g(x)$ into every x in $f(x)$, simplify algebraically, and then evaluate at the desired value.
- Inverse Function Definition — $f(g(x)) = x$ and $g(f(x)) = x$ Two functions are inverses of each other if composing them (in either order) returns the original input. We write f^{-1} for the inverse of f .
- Finding an Inverse Algebraically — Steps: 1) $y = f(x)$ 2) swap x and y 3) solve for y 4) $y = f^{-1}(x)$ Swapping x and y reverses the roles of input and output, which is exactly what an inverse does.
- Inverse from a Table or Graph — $f^{-1}(b) = a$ if $f(a) = b$ To find f^{-1} of a value, look for that value in the output column (or y -coordinate) and read off the corresponding input.
- Graphical Relationship — The graphs of f and f^{-1} are reflections of each other across the line $y = x$.

Watch Out: In $f(g(x))$, always simplify the INNER function's output (or expression) completely before substituting into the outer function — don't substitute term-by-term. ⚠️ $(x + 2)^2 \neq x^2 + 4$ — a squared binomial substituted into a composite function must be expanded with FOIL/the perfect-square rule, never distributed directly.

A function f is defined implicitly by the equation $2f(x) - 3x = 8$. Let $g(x) = x + 5$. If f^{-1} denotes the inverse of f , find the value of $g(f^{-1}(7))$.

Formula / Rule Used: Solve an implicit equation for $f(x)$ first, find its inverse algebraically (swap x and y , solve for y), then evaluate the composite in the given order.

Work through it, then advance for the full step-by-step solution.

Step 1: Solve the implicit equation for $f(x)$: $2f(x) - 3x = 8$, so $2f(x) = 3x + 8$, and $f(x) = (3x + 8)/2$.

Step 2: Find the inverse of f by replacing $f(x)$ with y : $y = (3x + 8)/2$.

Step 3: Swap x and y : $x = (3y + 8)/2$.

Step 4: Solve for y : $2x = 3y + 8$, so $3y = 2x - 8$, so $y = (2x - 8)/3$. This means $f^{-1}(x) = (2x - 8)/3$.

Step 5: Evaluate $f^{-1}(7)$: $f^{-1}(7) = (2(7) - 8)/3 = (14 - 8)/3 = 6/3 = 2$.

Step 6: Evaluate g at this result: $g(f^{-1}(7)) = g(2) = 2 + 5 = 7$.

Answer: $g(f^{-1}(7)) = 7$

Tip: When a function is given implicitly, isolate $f(x)$ completely first before doing anything else — you cannot find an inverse or compose functions from an equation that isn't solved for $f(x)$ yet.

Trap: The notation $f^{-1}(7)$ invites two classic errors here: treating it as $1/f(7)(a)$ reciprocal) or evaluating the functions in the wrong order by misreading which one is 'inside'.

Tempting Wrong Answers — and Exactly What Went Wrong

16/3

Reversed the composition order, computing $f^{-1}(g(7))$ instead of $g(f^{-1}(7))$ — first found $g(7) = 12$, then applied f^{-1} to that.

37/3

Made a sign error isolating y when finding the inverse, solving $x = (3y+8)/2$ as $y = (2x+8)/3$ instead of $y = (2x-8)/3$, then continued the rest of the problem with that wrong inverse.

39/2

Skipped finding the inverse entirely and evaluated $f(7)$ instead of $f^{-1}(7)$, using $f(7) = (3(7) + 8)/2 = 2\frac{1}{2}$ as the input to g .

Shortcut: Since g is just 'add 5', once you know $f^{-1}(7) = 2$ the final answer is simply $2 + 5$ — no need to write a separate composition line.

Desmos Tip: Graph $y = (3x + 8)/2$ and its reflection $y = (2x - 8)/3$ across $y=x$ to confirm they're inverses, then trace the second graph to $x=7$ to read off $f^{-1}(7) = 2$.

Core Rules — In Detail

- Vertical Shift — $y = f(x) + k$ Shifts the graph k units — up if $k > 0$, down if $k < 0$. Every y -value changes by k ; x -values are untouched.
- Horizontal Shift — $y = f(x + h)$ Shifts the graph h units LEFT if $h > 0$, and RIGHT if $h < 0$ — the opposite direction you'd expect from the sign.
- Reflections — $y = -f(x)$ reflects across the x -axis. $y = f(-x)$ reflects across the y -axis. Negating the whole output flips the graph vertically; negating the input flips it horizontally.
- Vertical Stretch/Compression — $y = a \cdot f(x)$ If $|a| > 1$, the graph stretches vertically (taller). If $0 < |a| < 1$, it compresses (shorter). If $a < 0$, it also reflects across the x -axis.
- Combined Transformation — $y = a \cdot f(x - h) + k$ Read transformations from the inside out: horizontal shift by h , vertical stretch/compression and reflection by a , then vertical shift by k .

Watch Out: Horizontal shifts are backwards from intuition: $f(x+2)$ shifts LEFT 2, not right — to keep the same output as the original at $x=0$, the new input must be 2 less. ⚠ When a problem gives you a transformed graph (like $y = f(x+2)+1$) and asks about the ORIGINAL $f(x)$, work backward: undo the transformations in reverse order.

A function f is defined only by the table below, for integer inputs from -4 to 5:

x : -4 -3 -2 -1 0 1 2 3 4 5

$f(x)$: 8 -3 0 1 2 4 9 1 6 -2

Let $g(x) = a \cdot f(x + 3) - 5$ for some constant a . If $g(1) = 7$, what is $g(-4)$?

Formula / Rule Used: Combined transformation $g(x) = a \cdot f(x+h)+k$: horizontal shift left by h ($h>0$), vertical stretch by a , vertical shift by k .

Work through it, then advance for the full step-by-step solution.

Step 1: Substitute $x = 1$ into $g(x) = a \cdot f(x+3) - 5$: $g(1) = a \cdot f(1+3) - 5 = a \cdot f(4) - 5$.

Step 2: From the table, $f(4) = 6$, so $g(1) = 6a - 5$.

Step 3: Set this equal to the given value: $6a - 5 = 7$.

Step 4: Solve for a : $6a = 12$, so $a = 2$.

Step 5: Now find $g(-4)$ using the same rule: $g(-4) = a \cdot f(-4+3) - 5 = 2 \cdot f(-1) - 5$.

Step 6: From the table, $f(-1) = 1$.

Step 7: Compute: $g(-4) = 2(1) - 5 = 2 - 5 = -3$.

Answer: $g(-4) = -3$

Tip: When a transformation formula has an unknown constant, first solve for that constant using the one given point, then re-apply the FULL rule (including the input shift) to the new input — don't skip straight to plugging the new x into f .

Trap: It's tempting to plug $x = -4$ directly into f from the table once a is known, forgetting that g still shifts the input by $+3$ first, which silently swaps $f(-1)$ for $f(-4)$ and wrecks the final answer.

Tempting Wrong Answers — and Exactly What Went Wrong

19

Treated $g(x) = a \cdot f(x) - 5$ without applying the +3 input shift at all, using $f(1)=4$ and $f(-4)=8$ instead of $f(4)$ and $f(-1)$, so a solved to 3 and the final value came out to $3(8)-5=19$.

7

Correctly found $a = 2$, but added 5 instead of subtracting it in the last step ($2(1)+5=7$), mixing up the sign of the vertical shift.

11

Correctly solved $a = 2$, but then forgot to re-apply the +3 shift when evaluating $g(-4)$, plugging $f(-4)=8$ directly instead of $f(-4+3)=f(-1)=1$, giving $2(8)-5=11$.

Shortcut: Once $a = 2$ is found, skip rewriting the whole equation again — just compute the shifted input ($-4+3 = -1$), read $f(-1)$ from the table, and plug it straight into $2(\cdot) - 5$.

Desmos Tip: Enter the table's x -values and $f(x)$ -values as two lists in Desmos, verify $a=2$ makes point $(1,7)$ satisfy g , then read g at the shifted input -1 to confirm -3 .

Core Rules — In Detail

- Polynomial Vocabulary — $a_n x^n + \dots + a_1 x + a_0$ Degree = highest exponent; leading coefficient = coefficient of the highest-degree term; like terms share the same variable AND exponent.
- Adding/Subtracting Polynomials — Combine only like terms, adding or subtracting their coefficients. When subtracting, distribute the negative sign to EVERY term of the second polynomial first.
- Distributive Property / FOIL — $a(b+c) = ab + ac$. $(a+b)(c+d) = ac+ad+bc+bd$ For binomials, FOIL (First, Outer, Inner, Last) is the distributive property applied twice; for larger polynomials, multiply every term in the first by every term in the second.
- Square of a Sum — $(a + b)^2 = a^2 + 2ab + b^2$ Square the first term, double the product of the two terms, square the last term.
- Square of a Difference — $(a - b)^2 = a^2 - 2ab + b^2$ Same pattern as the sum, but the middle term is negative.
- Difference of Squares — $(a + b)(a - b) = a^2 - b^2$ The middle terms cancel exactly — there is never a middle term in the expanded result.

Watch Out: $(x + 5)^2$ is NEVER $x^2 + 25$ — the middle term $2ab$ is real and is the single most common polynomial-operations mistake on the SAT. 
When subtracting one polynomial from another, every single term in the second polynomial changes sign — forgetting even one term is the most common subtraction error.



If $(x + p)^2 = x^2 + 26x + q$ for all real numbers x , where p and q are positive constants, what is the value of $q - p$?

Formula / Rule Used: Perfect square expansion: $(a + b)^2 = a^2 + 2ab + b^2$, matched coefficient by coefficient.

Work through it, then advance for the full step-by-step solution.

Step 1: Expand $(x + p)^2$ using the special product rule: $(x + p)^2 = x^2 + 2px + p^2$.

Step 2: Since this equals $x^2 + 26x + q$ for every x , the x -coefficients must match: $2p = 26$.

Step 3: Solve for p : $p = 13$.

Step 4: Match the constant terms: $q = p^2$.

Step 5: Substitute $p = 13$: $q = 13^2 = 169$.

Step 6: Compute $q - p = 169 - 13 = 156$.

Answer: $q - p = 156$

Tip: When a perfect-square expansion is set equal to a general trinomial, match the x -coefficient first to find the base constant, then square it to get the last term — don't try to read q off directly.

Trap: It's tempting to treat q as if it appears directly in the expansion (like $q = 26$, copying the x -coefficient) instead of recognizing that q is actually p^2 , which requires an extra squaring step after finding p .



Tempting Wrong Answers — and Exactly What Went Wrong

$q - p = 650$

Set $2p = 26$ but forgot to divide by 2, using $p = 26$ instead of 13, then computed $q = 26^2 = 676$, giving $q - p = 650$.

$q - p = -156$

Correctly found $p = 13$ and $q = 169$, but subtracted in the wrong order ($p - q$ instead of $q - p$), getting -156.

$q - p = 13$

Mistakenly used $q = 2p$ (mixing up the linear and squared terms of the identity) instead of $q = p^2$, getting $q = 26$ and $q - p = 26 - 13 = 13$.

Shortcut: Once $2p = 26$ gives $p = 13$, there's no need to re-expand anything symbolically — just square 13 directly to get $q = 169$.

Desmos Tip: Graph $y = (x + 13)^2$ and $y = x^2 + 26x + 169$ — the two curves should perfectly overlap, confirming $p = 13$ and $q = 169$ before computing the difference.

Core Rules — In Detail

- Factor Out the GCF First — Always check for a greatest common factor (of both the numbers and the variables — using the SMALLEST exponent of each variable present) before trying any other method.
- Factoring $x^2 + bx + c$ (leading coefficient 1) — Find r, s with $rs=c$ and $r+s=b$, then $(x+r)(x+s)$ List factor pairs of c and check which pair sums to b — track signs carefully when c is negative.
- The AC Method for $ax^2 + bx + c$ ($a \neq 1$) — Find factors of $a \times c$ that sum to b , rewrite bx as their sum, then factor by grouping. This turns a hard trinomial into two easier grouping steps.
- Difference of Squares — $a^2 - b^2 = (a - b)(a + b)$ Recognizable whenever two perfect squares are subtracted — check twice, since $a^4 - 81$ factors as a difference of squares AND then factors again.
- Perfect Square Trinomial — $a^2 \pm 2ab + b^2 = (a \pm b)^2$ Check: is the first term a perfect square, the last term a perfect square, and the middle term exactly twice their product?
- Factoring by Grouping (Cubics) — Group the first two and last two terms, factor each group separately, then factor out the common binomial that remains.
- Multiplicity of Roots — The exponent on a factor is that root's multiplicity. Odd multiplicity: the graph crosses the x-axis. Even multiplicity: the graph touches the x-axis and turns back.

Watch Out: A polynomial isn't 'fully factored' just because you found one difference of squares — always check whether either resulting factor can be factored again. ⚠️ $x^2 + 9$ (a sum of squares) does NOT factor over the real numbers — don't force a factorization that doesn't exist.

The equation $12x^2 - 17x = 40$ has two real solutions. What is the positive difference between them?

Formula / Rule Used: Rearrange to standard form $ax^2 + bx + c = 0$, then factor by the AC method (grouping) when a is not 1.

Work through it, then advance for the full step-by-step solution.

Step 1: Rearrange to standard form: $12x^2 - 17x - 40 = 0$.

Step 2: For the AC method, multiply a and c: $12 \times (-40) = -480$.

Step 3: Find two numbers that multiply to -480 and add to -17: -32 and 15 work, since $(-32)(15) = -480$ and $-32 + 15 = -17$.

Step 4: Rewrite the middle term and group: $12x^2 - 32x + 15x - 40 = 0$.

Step 5: Factor each pair: $4x(3x - 8) + 5(3x - 8) = 0$.

Step 6: Factor out the common binomial: $(4x + 5)(3x - 8) = 0$.

Step 7: Solve each factor: $4x + 5 = 0$ gives $x = -5/4$; $3x - 8 = 0$ gives $x = 8/3$.

Step 8: Find the positive difference: $8/3 - (-5/4) = 32/12 + 15/12 = 47/12$.

Answer: 47/12

Tip: Before applying the AC method, always move every term to one side so the equation reads $ax^2 + bx + c = 0$ -- factoring a rearranged equation directly produces the wrong factors.

Trap: A student who factors $12x^2 - 17x = 40$ without rearranging might factor the left side alone as $x(12x-17)$ and effectively solve $12x^2 - 17x = 0$, 'losing' the 40 and getting the false solutions $x=0$ and $x = 17/12$.

Tempting Wrong Answers — and Exactly What Went Wrong

17/12

Forgot to move the 40 to the left side and instead factored $12x^2 - 17x = x(12x - 17) = 0$, getting the incorrect solutions $x = 0$ and $x = \frac{17}{12}$.

17/3

Chose the wrong number pair in the AC method (8 and -60, which multiply to -480 but add to -52, not -17), leading to incorrect binomials $(4x-20)(3x+2)$ and roots $x=5$ and $x=-2/3$.

13/7

Found the correct roots $8/3$ and $-5/4$ but added the fractions incorrectly by combining numerators and denominators separately $(8-(-5))/(3+4)$ instead of finding a common denominator.

Shortcut: Skip full factoring for the difference: $|r_1 - r_2| = \sqrt{(b^2 - 4ac)/a} = \sqrt{((-17)^2 - 4(12)(-40))/12} = \sqrt{2209/12} = \frac{47}{12}$, directly.

Desmos Tip: Graph $y = 12x^2 - 17x - 40$ and use the calculator's intersection/root feature to find both x-intercepts, then subtract them directly.

Core Rules — In Detail

- Dividing by a Monomial — $(a+b+c)/d = a/d + b/d + c/d$ Split into separate fractions and simplify each term individually.
- Polynomial Long Division — Divide leading terms to get the next quotient term, multiply that term by the whole divisor, subtract, bring down the next term, and repeat until the remainder's degree is less than the divisor's.
- Simplifying Rational Expressions — Factor the numerator and denominator completely, then cancel any common factors — the simplified expression excludes whatever values made the ORIGINAL denominator zero.
- Multiplying/Dividing Rational Expressions — Factor everything first, then cancel common factors across the whole product. For division, multiply by the reciprocal of the second fraction.
- Adding/Subtracting Rational Expressions — Find the least common denominator (LCD), rewrite each fraction over it, then combine the numerators.
- Solving Rational Equations — Multiply every term by the LCD to clear all denominators, solve the resulting equation, then discard any solution that makes an original denominator zero (extraneous).
- Complex Fractions — Either combine the numerator and denominator into single fractions and then divide, or multiply the entire complex fraction by the LCD of all the inner fractions.

Watch Out: Canceling a factor changes the domain unless you track it — the simplified expression is only equivalent to the original for x-values that didn't make the original denominator zero. ⚠ Every rational-equation solution must be checked against every original denominator, not just the one that produced it — an extraneous solution can come from any denominator in the problem.

When the polynomial $3x^3 + ax^2 - 7x + 4$ is divided by $(x - 2)$, the quotient is $3x^2 + 5x + 3$ and the remainder is 10. Find the value of a .

Formula / Rule Used: Division Algorithm: Dividend = Divisor x Quotient + Remainder; equivalently, Remainder Theorem: dividing by $(x-k)$ leaves remainder $P(k)$.

Work through it, then advance for the full step-by-step solution.

Step 1: Recall the division algorithm: Dividend = Divisor x Quotient + Remainder.

Step 2: Recognize the faster path: dividing by $(x-2)$ means the Remainder Theorem applies, so $P(2)$ equals the remainder, 10.

Step 3: Substitute $x=2$ into $3x^3 + ax^2 - 7x + 4$: $3(2)^3 + a(2)^2 - 7(2) + 4 = 3(8) + 4a - 14 + 4$.

Step 4: Simplify the constants: $24 - 14 + 4 = 14$, so $P(2) = 14 + 4a$.

Step 5: Set this equal to the given remainder: $14 + 4a = 10$.

Step 6: Solve: $4a = -4$, so $a = -1$.

Step 7: Check by expansion: $(x - 2)(3x^2 + 5x + 3) + 10 = 3x^3 - x^2 - 7x + 4$, which matches $3x^3 + ax^2 - 7x + 4$ exactly when $a=-1$.

Answer: $a = -1$

Tip: When a problem gives you a quotient AND a remainder and asks for a missing coefficient in the dividend, plug the divisor's root into the dividend (Remainder Theorem) instead of expanding divisor times quotient by hand.

Trap: Students who don't trust the Remainder Theorem often expand $(x - 2)(3x^2 + 5x + 3)$ term by term, and a sign slip in that expansion (especially with the -2 times $5x$ term) is far more likely than the one-line substitution.

Tempting Wrong Answers — and Exactly What Went Wrong

a = -8

Computed $-7(2)$ as $+14$ instead of -14 (dropped the negative sign), giving $24+4a+14+4=10$ and solving $4a=-32$.

a = -3.5

Confused the Remainder Theorem with the Factor Theorem and set $P(2)$ equal to 0 (as if the divisor left no remainder) instead of equal to the given remainder 10.

a = 0.5

Miscomputed 2^3 as $2 \times 3 = 6$ instead of $2 \times 2 \times 2 = 8$, corrupting the first term of $P(2)$ to 18 instead of 24.

Shortcut: The Remainder Theorem already IS the shortcut here — there's no need to ever multiply out $(x - 2)(3x^2 + 5x + 3)$; a one-line substitution of $x=2$ gets the answer directly.

Desmos Tip: Graph $y = 3x^3 - x^2 - 7x + 4$ (using $a=-1$) and check that the y-value at $x=2$ reads exactly 10, matching the given remainder.

Core Rules — In Detail

- Solving by Factoring — $ax^2 + bx + c = 0 \rightarrow \text{factor} \rightarrow \text{set each factor to 0}$ Fastest method whenever the trinomial factors nicely; always check that the equation is set equal to zero first.
- Solving by Taking Square Roots — If $x^2 = k$, then $x = \pm\sqrt{k}$. Use when the equation is (or can be isolated to) a squared term equal to a constant — never forget the $\pm$ sign.
- Solving by Completing the Square — Move the constant to the other side, take half the x-coefficient, square it, add that to both sides, then factor the left side as a perfect square.
- The Quadratic Formula — $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$ Works for ANY quadratic equation, especially when factoring is difficult or impossible — always identify a, b, and c carefully first.
- The Discriminant — $D = b^2 - 4ac$ $D > 0$: two distinct real solutions. $D = 0$: exactly one real solution (a repeated root). $D < 0$: no real solutions.

Watch Out: Forgetting the $\pm$ when taking a square root loses half of the valid solutions — this is one of the most common quadratic-solving errors.

! Before using the discriminant or quadratic formula, the equation MUST be in standard form $ax^2+bx+c=0$ — rearrange first if it isn't.



The equation $3x^2 - kx + 12 = 0$, where k is a positive constant, has a discriminant equal to 25. What is the value of k ?

Formula / Rule Used: Discriminant $D = b^2 - 4ac$.

Work through it, then advance for the full step-by-step solution.

Step 1: Identify $a = 3$, $b = -k$, and $c = 12$ in the equation $3x^2 - kx + 12 = 0$.

Step 2: Write the discriminant symbolically: $D = b^2 - 4ac = (-k)^2 - 4(3)(12)$.

Step 3: Simplify: $D = k^2 - 144$.

Step 4: Set this equal to the given discriminant value: $k^2 - 144 = 25$.

Step 5: Solve for k : $k^2 = 169$.

Step 6: Take the square root of both sides, remembering both signs: $k = \pm 13$.

Step 7: Since the problem states k is positive, discard $k = -13$ and keep $k = 13$.

Answer: $k = 13$

Tip: Write the discriminant expression symbolically first (in terms of the unknown), then set it equal to the target value and solve like any other equation.

Trap: A rushed student who solves $k^2 = 169$ might report both roots or grab the negative one without noticing the problem restricts k to positive values, since -13 is a perfectly valid algebraic solution to $k^2 = 169$.

Tempting Wrong Answers — and Exactly What Went Wrong

k = -13

Solved $k^2 = 169$ correctly but ignored the problem's stated restriction that k is positive, reporting the negative root instead.

No real value of k works

Used $D = b^2 + 4ac$ instead of $D = b^2 - 4ac$ (sign error), getting $k^2 + 144 = 25$, i.e. $k^2 = -119$, which incorrectly appears to have no real solution.

k = 169

Forgot to square $b = -k$ when writing the discriminant, using $D = k - 4ac$ instead of $D = k^2 - 4ac$, so $k - 144 = 25$ gave $k = 169$.

Shortcut: Since only the positive value of k is requested, you can skip writing ' $\pm$ ' altogether and just take the positive square root of 169 once you reach $k^2 = 169$.

Desmos Tip: Graph $y = 3x^2 - 13x + 12$ on Desmos and confirm it crosses the x-axis at two distinct points, consistent with a positive discriminant.

Core Rules — In Detail

- Standard Form — $y = ax^2 + bx + c$ c is the y-intercept; the vertex's x-coordinate is $x_v = -b/2a$.
- Factored Form — $y = a(x - p)(x - q)$ p and q are the x-intercepts (roots); the vertex's x-coordinate is the average of the roots, $(p+q)/2$.
- Vertex Form — $y = a(x - h)^2 + k$ (h, k) is the vertex directly — the minimum point if $a > 0$, the maximum point if $a < 0$.
- Completing the Square to Reach Vertex Form — Factor a out of the first two terms, add and subtract $(\text{half the new x-coefficient})^2$ inside, then factor the resulting perfect square.
- Sum and Product of Roots (Vieta's Formulas) — Sum = $-b/a$. Product = c/a . Lets you find a combination of the roots without ever solving the equation directly.
- Quadratic Word Problems — Translate the scenario into a quadratic equation (area = length $\times$ width, height = $-16t^2 + v_0t + h_0$, etc.), solve, and discard any solution that isn't physically valid (negative length, negative time).

Watch Out: A parabola opening downward ($a < 0$) has a MAXIMUM at its vertex, not a minimum — always check the sign of a before describing the vertex. ⚠ In a word problem, always discard algebraically valid solutions that make no real-world sense (negative width, negative time before an event starts).

A quadratic function has vertex $(3, -16)$. Its graph crosses the x -axis at two points, and the product of the two x -intercepts is -7 . If the function is written as $y = a(x - 3)^2 - 16$, what is the value of a ?

Formula / Rule Used: Vieta's formulas: sum of roots $= -B/A$, product of roots $= C/A$, applied after expanding vertex form to standard form.

Work through it, then advance for the full step-by-step solution.

Step 1: Expand the vertex form: $y = a(x - 3)^2 - 16 = a(x^2 - 6x + 9) - 16 = ax^2 - 6ax + (9a - 16)$.

Step 2: Identify the standard-form coefficients: $A = a$, $B = -6a$, $C = 9a - 16$.

Step 3: Apply Vieta's product-of-roots formula: $\text{product} = C/A = (9a-16)/a$.

Step 4: Set this equal to the given product, -7: $(9a-16)/a = -7$.

Step 5: Multiply both sides by a : $9a - 16 = -7a$.

Step 6: Solve: $9a + 7a = 16$, so $16a = 16$, and $a = 1$.

Answer: $a = 1$

Tip: Whenever a parabola is given in vertex form but the question refers to its roots (sum or product), expand to standard form first so you can apply Vieta's formulas directly on A, B, and C.

Trap: It's tempting to treat the product of roots as $-C/A$ (mirroring the sum formula's minus sign) instead of the correct C/A , which produces $a = 8$ instead of $a = 1$.

Tempting Wrong Answers — and Exactly What Went Wrong

8

Used $-C/A$ instead of C/A for the product of roots (mixing up the sign pattern from the sum formula), solving $-(9a-16)/a = -7$ to get $a = 8$.

-1

Made a sign error moving terms, writing $9a + 16 = -7a$ instead of $9a - 16 = -7a$, which gives $16a = -16$ and $a = -1$.

-8

Expanded $(x - 3)^2$ incorrectly as $x^2 - 9$ (misapplying a difference-of-squares pattern instead of squaring the binomial), which changed C and produced $a = -8$ instead of $a = 1$.

Shortcut: Since the vertex's x -coordinate already tells you the roots' sum is $2(3) = 6$, you can skip straight to setting up the product = C/A equation without separately re-deriving or checking the sum.

Desmos Tip: Graph $y = (x - 3)^2 - 16$ (using $a = 1$) and confirm the x -intercepts are at $x = -1$ and $x = 7$, whose product $(-1)(7) = -7$ matches the given condition.

Core Rules — In Detail

- Radical-Exponent Conversion — $\sqrt[n]{a^m} = a^{m/n}$ The denominator of the fractional exponent is the root (index); the numerator is the power.
- Properties of Radicals — $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ $\sqrt[n]{a/b} = \sqrt[n]{a} / \sqrt[n]{b}$ Valid for non-negative radicands when the index is even; valid for all real radicands when the index is odd.
- Simplifying Radicals — Factor out the largest perfect power (square, cube, etc.) hiding inside the radicand and pull it out.
- Adding/Subtracting Radicals — Only radicals with the SAME index and SAME radicand can be combined, exactly like combining like terms — simplify each radical first, since disguised matches are common.
- Rationalizing Denominators — Single term: multiply by the radical. Binomial: multiply by the conjugate. The conjugate of $a+\sqrt{b}$ is $a-\sqrt{b}$; multiplying a binomial radical denominator by its conjugate eliminates the radical via the difference-of-squares pattern.
- Exponent Rules with Rational Exponents — $a^{m/n} \cdot a^{p/n} = a^{(m+p)/n}$ The product, quotient, and power rules for integer exponents apply identically to fractional exponents.

Watch Out: An even-index root of a negative radicand is not a real number — $\sqrt{-4}$ has no real value, but cube roots of negative numbers ARE real: $\text{cbrt}(-8) = -2$. ⚠ Before adding or subtracting radicals, simplify each one first — $\sqrt{45} + \sqrt{20}$ look unrelated until simplified to $3\sqrt{5} + 2\sqrt{5}$.

Simplify the expression $[(a^{\frac{3}{4}} \cdot b^{-\frac{3}{4}})^3] / (a^{\frac{3}{4}} \cdot b^{-1})$, then evaluate the result for $a = 4$ and $b = 9$.

Formula / Rule Used: Rational exponent rules: $(x^m)^n = x^{m \cdot n}$; $x^m / x^n = x^{m-n}$.

Work through it, then advance for the full step-by-step solution.

Step 1: Apply the power rule to the numerator: $(a^{\frac{2}{3}})^3 = a^{\frac{2}{3} \cdot 3} = a^2$, and $(b^{-\frac{1}{3}})^3 = b^{-1}$. The numerator becomes $a^2 \cdot b^{-1}$.

Step 2: Divide by the denominator's a-term using the quotient rule: $a^2/a^{\frac{1}{2}} = a^{2 - \frac{1}{2}} = a^{\frac{3}{2}}$.

Step 3: Divide by the denominator's b-term: $b^{-\frac{3}{2}}/b^{-1} = b^{-\frac{3}{2} - (-1)} = b^{-\frac{1}{2}}$.

Step 4: The fully simplified expression is $a^{\frac{3}{2}} \cdot b^{-\frac{1}{2}}$, which is the same as $a^{\frac{3}{2}}/b^{\frac{1}{2}}$.

Step 5: Evaluate $a^{\frac{3}{2}}$ at $a = 4$: $a^{\frac{3}{2}} = (a^{\frac{1}{2}})^3 = (\sqrt{4})^3 = 2^3 = 8$.

Step 6: Evaluate $b^{\frac{1}{2}}$ at $b = 9$: $\sqrt{9} = 3$.

Step 7: Combine the results: $8/3$.

Answer: $8/3$

Tip: Fully simplify the exponents symbolically first — combine everything into a single power of a and a single power of b — before substituting numbers; plugging in early forces you to compute ugly fractional powers like $4^{\frac{3}{2}}$ instead of clean ones like 4^3 .

Trap: A student who substitutes $a = 4$ and $b = 9$ immediately, before simplifying the exponent algebra, ends up needing to compute things like $4^{\frac{3}{2}}$ and $9^{-\frac{1}{2}}$ directly inside a cubed expression and often mishandles the negative exponent on b, landing on $\frac{3}{8}$ instead of $\frac{8}{3}$.

Tempting Wrong Answers — and Exactly What Went Wrong

3/8 Simplified the exponents correctly but mishandled the negative sign on $b^{-\frac{3}{2}}$, inverting the final fraction so b ended up in the numerator instead of the denominator.

24 Correctly got $a^{\frac{2}{3}} = 8$ but treated $b^{-\frac{3}{2}}$ as $b^{\frac{3}{2}}$ (dropped the negative sign entirely), computing $8 * 3 = 24$ instead of $\frac{8}{3}$.

8/9 Correctly simplified a's exponent to $a^{\frac{2}{3}} = 8$ but never finished subtracting the b exponents down to $-\frac{1}{2}$, plugging in $b = 9$ as if its exponent were still -1, giving $\frac{8}{9}$.

Shortcut: Track the exponent on a and the exponent on b as two separate running totals (starting at $\frac{2}{3} * 3 = 2$ and $-\frac{1}{2} * 3 = -\frac{3}{2}$) and subtract the denominator's exponents from each as you go, instead of writing out the full expanded expression with both variables tangled together.

Desmos Tip: Enter $(a^{\frac{2}{3}} \cdot b^{-\frac{3}{2}}) / (a^{\frac{2}{3}} \cdot b^{-1})$ into Desmos with $a = 4$ and $b = 9$ typed in directly to numerically confirm the result is $\frac{8}{3}$ (about 2.667).

Core Rules — In Detail

- General Exponential Form — $y = a \cdot b^x$ a is the initial value (value at $x=0$); b is the growth/decay factor. If $b>1$, growth; if $0<b<1$, decay.
- Percent Change Connection — Growth: $A = P(1+r)^t$. Decay: $A = P(1-r)^t$. r is the percent rate as a decimal; the base b in the general form equals $(1+r)$ for growth or $(1-r)$ for decay.
- Key Features of the Basic Graph — y-intercept $(0, a)$; horizontal asymptote $y = 0$. Domain is all real numbers; range is $y > 0$ for a positive initial value — the graph never touches its asymptote.
- Solving with the Same-Base Method — If $a^m = a^n$ (same base a), then $m = n$. Rewrite both sides with the same base when possible (e.g. 9 and 27 both as powers of 3), then set the exponents equal and solve.
- Transformations of Exponential Graphs — $y = b^x + k$ shifts the asymptote to $y = k$. $y = b^{x-h}$ shifts horizontally. A vertical shift changes the horizontal asymptote itself, not just the graph's position.

Watch Out: A vertical shift on an exponential graph moves the asymptote too — $y = 2^x + 3$ has asymptote $y = 3$, not $y = 0$. ⚠ When rewriting exponential equations to a common base, remember that raising a power to a power MULTIPLIES exponents — a very common slip when converting bases like $9 = 3^2$.

A tech startup based in Thessaloniki has revenue that grows exponentially over time. Company records show that revenue was €120,000 in year 2 after launch and €172,800 in year 4 after launch. The company currently employs 45 people. Assuming the same exponential growth pattern continues, what will the company's revenue be in year 7 after launch, to the nearest euro?

Formula / Rule Used: Exponential growth model $A = a \cdot b^x$; divide two known outputs to isolate a power of b .

Work through it, then advance for the full step-by-step solution.

Step 1: Let $R(t) = a \cdot b^t$ model revenue t years after launch, where a is the initial revenue and b is the annual growth factor.

Step 2: Divide the year-4 revenue by the year-2 revenue to eliminate a : $(a \cdot b^4) / (a \cdot b^2) = b^2 = 172,800 / 120,000 = 1.44$.

Step 3: Solve for b : $b = \sqrt{1.44} = 1.2$, so revenue grows by 20% each year.

Step 4: Rather than solving for a , project forward directly from year 4: since $7 - 4 = 3$ years pass, $R(7) = R(4) \cdot b^3$.

Step 5: Compute b^3 : $1.2^3 = 1.2 \cdot 1.2 \cdot 1.2 = 1.728$.

Step 6: Multiply: $R(7) = 172,800 \times 1.728 = 298,598.4$.

Step 7: Round to the nearest euro: $R(7) \approx \text{€}298,598$. (The fact that the company has 45 employees plays no role in the calculation.)

Answer: €298,598

Tip: When two outputs of an exponential model are given at two different input values, divide them to isolate a power of b — never average or subtract the two values.

Trap: A rushed student takes 1.44 itself as the annual growth factor instead of recognizing it as b^2 and taking its square root, which compounds the growth far too fast over the remaining years.

Tempting Wrong Answers — and Exactly What Went Wrong

≈€743,046

Used 1.44 directly as the one-year growth factor instead of taking its square root to isolate the true annual multiplier $b = 1.2$, so every remaining year compounded too aggressively.

€252,000

Modeled the growth as linear, adding a flat $(172,800 - 120,000)/2 = 26,400$ per year, instead of recognizing that revenue changes multiplicatively, not additively.

€358,318

Correctly found $b = 1.2$ and used $R(4)$ as the base, but miscounted the gap from year 4 to year 7 as 4 years instead of 3, computing $R(4) \cdot 1.2^4$ instead of $R(4) \cdot 1.2^3$.

Shortcut: Once b is found, project forward from the later known data point ($R(4)$) using b raised to the remaining number of years, instead of solving for the initial value a first and restarting from year 0.

Desmos Tip: Plot the points (2, 120000) and (4, 172800), fit $y = a \cdot b^x$ by adjusting a and b until the curve passes through both, then trace to $x = 7$ to read the projected revenue.

Core Rules — In Detail

- Solving Radical Equations — Isolate the radical on one side, raise both sides to the power matching the index, solve the resulting equation, then plug every solution back into the ORIGINAL equation.
- Extraneous Solutions from Radicals — Raising both sides to an EVEN power can introduce solutions that don't satisfy the original equation — always check. Odd-index roots (cube roots) never create this problem.
- Radical Equations with Two Radicals — Isolate one radical, square, then isolate the remaining radical and square again if needed.
- Solving Rational Equations — Multiply every term by the least common denominator (LCD) to clear all fractions at once, solve the resulting polynomial equation, then discard any solution making an original denominator zero.
- Complex Fractions — Multiply the entire complex fraction (top and bottom) by the LCD of every inner fraction to clear all the small fractions in one step.

Watch Out: 'Solve, then check' is not optional for radical equations raised to an even power — a fully correct algebraic solution can still be extraneous. ⚠ In a rational equation, a value is extraneous if it zeroes out ANY original denominator — check every denominator in the problem, not just the most obvious one.

Find all real values of x that satisfy: $\sqrt{2x + 9} - \sqrt{x + 4} = 1$.

Formula / Rule Used: For equations with two radicals, isolate one radical, square, isolate the remaining radical, then square again.

Work through it, then advance for the full step-by-step solution.



Step 1: Isolate one radical by moving the other to the right side: $\sqrt{2x + 9} = 1 + \sqrt{x + 4}$.

Step 2: Square both sides: $2x + 9 = 1 + 2\sqrt{x + 4} + (x + 4) = x + 5 + 2\sqrt{x + 4}$.

Step 3: Isolate the remaining radical: $2x + 9 - x - 5 = 2\sqrt{x + 4}$, so $x + 4 = 2\sqrt{x + 4}$.

Step 4: Square both sides again: $(x + 4)^2 = 4(x + 4)$.

Step 5: Move everything to one side and factor out $(x + 4)$: $(x + 4)^2 - 4(x + 4) = (x + 4)$.

Step 6: So $x = -4$ or $x = 0$.

Step 7: Check $x = -4$ in the original equation: $\sqrt{2(-4) + 9} - \sqrt{-4 + 4} = \sqrt{1} - \sqrt{0} = 1 - 0 = 1$. Valid.

Step 8: Check $x = 0$ in the original equation: $\sqrt{9} - \sqrt{4} = 3 - 2 = 1$. Also valid.

Step 9: Both candidates check out, so both are genuine solutions.

Answer: $x = -4$ or $x = 0$

Tip: With two radicals, isolate ONE of them before the first squaring — never square while both radicals sit on the same side, or the cross-term won't cancel cleanly.

Trap: A student who squares the original equation as $(\sqrt{2x + 9} - \sqrt{x + 4})^2 = 1$ without isolating first gets a messier cross term involving $\sqrt{(2x + 9)(x + 4)}$; moving pieces around from there is where an extra sign flip usually sneaks in.

Tempting Wrong Answers — and Exactly What Went Wrong

$x = 0$

Found both roots from the second squaring but assumed (without checking) that $x = -4$ must be extraneous, since two-radical problems 'usually' produce one bad root, and never actually tested it.

$x = 4$

Simplified $(x + 4) - 4$ incorrectly as $4 - x$ instead of x , giving $(x + 4)(4 - x) = 0$ and roots $x = -4$ or $x = 4$, then reported the unfamiliar-looking root $x = 4$ instead of the correct $x = 0$.

$x = -4$

Stopped after confirming the first valid check on $x = -4$ and never went back to test whether $x = 0$ was also a valid solution worth reporting.

Shortcut: Once you reach $x + 4 = 2\sqrt{x + 4}$, let $u = \sqrt{x + 4}$: the equation becomes $u^2 = 2u$, so $u = 0$ or $u = 2$, and converting back to $x(x = u^2 - 4)$ is one quick line for each.

Desmos Tip: Graph $y = \sqrt{2x + 9} - \sqrt{x + 4}$ and $y = 1$, then find both intersection points — the calculator shows two crossings, at $x = -4$ and $x = 0$, confirming neither is extraneous here.